Daniel Molitor

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Education

Cornell University 2022 – 2027 (Expected)
PhD in Information Science (Minor in Statistics)

Bethel University 2016 – 2019
B.A. in Mathematics and Economics

Publications

Anytime-Valid Inference in Adaptive Experiments: Covariate Adjustment and Balanced Power with Samantha Gold. *arXiv*, 2025.

Data-Adaptive Experimentation to Find Contexts with the Most and Least Discrimination with Jennah Gosciak and Ian Lundberg. *SocArXiv*, 2025.

The Causal Effect of Parent Occupation on Child Occupation: A Multivalued Treatment with Positivity Constraints with Jennie Brand and Ian Lundberg. *Sociological Methods & Research*, 2025.

Estimating Value-added Returns to Labor Training Programs with Causal Machine Learning with Mintaka Angell et al. *OSF Preprints*, 2021.

Delivering Unemployment Assistance in Times of Crisis: Scalable Cloud Solutions Can Keep Essential Government Programs Running and Supporting Those in Need with Mintaka Angell et al. *Digital Government: Research and Practice*, 2021, vol. 2, no. 1, pp. 1–11.

Invited Conferences

2026: IC2S2, American Causal Inference Conference

2025: INFORMS Annual Meeting, International Conference on Computational Social Science (IC2S2), American Causal Inference Conference

2024: Conference on Digital Experimentation @ MIT (CODE@MIT), American Sociological Association Annual Meeting, American Causal Inference Conference

2023: American Sociological Association Annual Meeting

Professional Experience

Google Current
Quantitative UX Research Intern

- **Supervisors**: Mackenzie Sunday, Zheng Ma

Philadelphia Phillies 2025
Quantitative Analyst Intern – Predictive Modeling Group

- **Supervisors**: Chris Adams, Patrick McFarlane
- **Description**: Built predictive models from complex internal datasets to generate defensive projections and assess player defensive performance.

Research Improving People's Lives 2021 – 2022
Pre-Doctoral Fellow

- **Research Topics**: career recommendations with large-scale administrative data, cloud solutions for state governments,
- **Research Supervisors**: Niall Keleher, Karen Shen, Justine Hastings
- **Description**: Leveraged causal and predictive approaches to derive occupational ROI metrics and deliver personalized career recommendations to jobseekers across several states, including RI, WA, and VA.

National Bureau of Economic Research 2019 – 2021

Pre-Doctoral Fellow

- **Research Topics**: labor economics, career transitions, applied causal machine learning, labor training program evaluation
- **Research Supervisors**: Justine Hastings, Mark Howison
- **Description**: Applied causal machine learning methods to large-scale administrative datasets to facilitate a causal evaluation of labor-training programs in RI.

Teaching

Cornell University

(Teaching Assistant) *INFO 3370: Studying Social Inequality with Data Science*, Prof. Ian Lundberg Spring 2024

(Teaching Assistant) *STSCI / INFO / ILRST 3900: Causal Inference*, Prof. Ian Lundberg & Prof. Sam Wang Fall 2023

Honors, Awards, and Grants

- AWS Cloud Computing Grant ("Data-Adaptive Experiments to Discover Discrimination in Context"), 2024
- 2023-2024 Outstanding Teaching Awards – Cornell Information Science
- National Science Foundation Graduate Research Fellowship
- NBER Pre-doctoral Fellow

Software (I've developed)

tugboat: Quickly turn your analysis directory into a Docker image and Binder-compatible project.

jetty: Execute R functions, code snippets, and scripts within the context of a Docker container.

bolasso: Implements model consistent penalized regression estimation through the bootstrap.

fulgur: Estimate a range of linear models on extremely large (out-of-core) datasets.

modeltuning: A light-weight, intuitive framework for common model selection and parameter tuning utilities.

pyssed: Robust and precise anytime-valid causal inference in adaptive experiments.

Additional Information

Technical Skills: Python, R, Docker, Git, SQL, Linux, Quarto, AWS, Shiny, NextFlow, SCons, MLFlow, Rust

Domain Experience: experimental design, causal inference, adaptive experimentation, user experience research, machine learning